



A digital watermark embedded in an image

The data is first compressed, encrypted and then hidden in the MP3 bit stream. An outsider, however, can attack it by uncompressing and recompressing the bitstream, deleting the hidden information with severe loss to quality.

Messages in video: Embedding information into multimedia data has gained increasing attention lately. The method of encryption is the same as in audio steganography. Video files are generally very good carrier files since they have a lot of irrelevant bits.

Steganography and watermarking

Steganography also enables an evolved form of watermarking. Usually, covert channels in the carrier file carry hidden data unrelated to the content in which it is embedded, but digital watermarking holds information about its carrier medium and is broadly defined as an imperceptible insertion of information such as a number or text into a multimedia file through slight data modification. This process has gained huge acclaim from the media for enabling copyright for their products.

The Law Effect

The World Intellectual Property Organization (WIPO) has drafted a treaty, which requires countries to penalise those who remove digital watermarks. The Digital Millennium Copyright Act (DMCA) seeks to prevent any individual from forging or removing a watermark, or from creating or publishing about hardware or software that is capable of doing so. New legislation now pending in the US Congress will, if it is passed, specifically penalise those who alter watermarks. Thus, you could not legally create a home recording on a player restricted to playing for example, only Disney titles or only titles for which a fee had been paid to the player's manufacturer or a consortium of recording companies.

Video steganography is more suited to avoiding piracy and is mostly used for digital watermarking. The owner's data is embedded either into the uncoded video or into the MPEG-2 bitstream and can be later retrieved from the decoded video. Although an existing MPEG-2 bitstream is partly altered, it does not visibly distort the original video information.

Many commercial graphics programs allow artistes to embed their names as well as copyright notices within images. Sometimes the concealed message may consist of a single bit that simply says 'That's done by me!' This is sufficient to accomplish the desired purpose—a recording or playback device can detect this watermark and refuse to function, disabling access to the data if the user doesn't have the authorisation.

There are two types of digital watermarks—robust and fragile. A robust watermark is embedded in the file in such a way that even if the file is later transformed, the watermark will not be removed. A robust digital watermark is a concealed message that identifies the source of the data. It is called robust because it is designed to survive modifications to the data that result from resizing, cropping, or photographing an image to capture only the part of the image that doesn't have the watermark, or even from re-sampling or making an analog recording of an audio stream. Its application includes copyright protection, labelling, monitoring, tamper proofing and conditional access.

On the other hand, fragile digital watermarks are similar to fragile analog watermarks—if the data is altered or copied inexactly, the watermark is corrupted. If changes are made to a file containing a fragile digital watermark, the originator of the watermark will be able to detect and identify the areas where the alterations have been made and maybe even determine what the data was before modification. This scheme serves at proving the authenticity of the data.

For ensuring the integrity of data, digital signatures are preferred but fragile digital watermarking can detect data tampering without alerting the culprit. Compatible players refuse to play content that does not bear a valid watermark.

All part of one family

So how is steganography different from encryption? While cryptography protects the content of messages (their meaning), steganography conceals their very existence. In encryption, the reader can see

It's a Secret!

Plenty of free and shareware steganography software can be found on the Net. A list of such software can be obtained at <http://www.cotse.com/tools/stega.htm> or http://www.cl.cam.ac.uk/~fapp2/steganography/stego_soft.html. We are using a software called S-Tools 4 to hide one image within another. The receiver would need the same software to decode it and would also need to know the passphrase and the encryption algorithm.

STEP 1 Start S-Tools and Windows Explorer, using the latter as a drag-and-drop interface for the software.

STEP 2 Drag and drop the image to be used as the carrier file from the Explorer onto the Actions window in S-Tools.



The carrier file

STEP 3 Drag and drop the data file on the carrier file.



The data file to be hidden within the carrier file

STEP 4 You will be prompted to secure the process using a passphrase and an encryption algorithm. Convey these to the receiver too.



Securing the data file through encryption

STEP 5 The hidden data file is ready. Once the recipient receives it, he or she can just right-click on the file and choose Reveal. To extract it, right-click on the image and save it in the destination folder.



Extracting the hidden file